

0/24 Questions Answered

hw9-regression-mle-gorillas

Student Name

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Q1

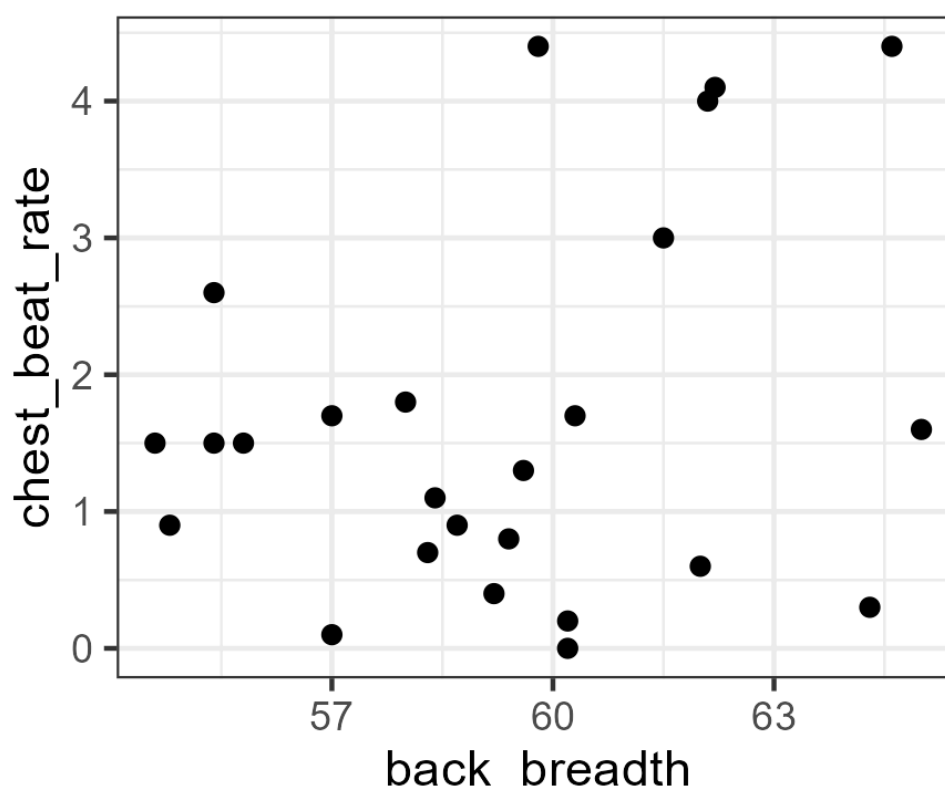
10 Points

Note: There are three options for homework 9, that differ by data topic, and you only need to complete one of the three assignments.

Acoustic signals of animals have been found to be key for facilitating assessment of rivals and in the choice of mates. Chest beating, where the animal stands on two legs and pounds on their chest to make a drumming sound, is a common characteristic of gorillas, especially male gorillas. This drumming can even be heard over half a mile away. We are interested in studying if we can use a gorillas size to learn about its chest beating behaviors, since they are related to dominance.

We have data on chest beating rate (per 10 hours),

`chest_beat_rate`, and maximum distance across the shoulders (cm), `back_breadth` for 25 wild male mountain gorillas in Rwanda.



First, install the R packages if you haven't.

```
install.packages(c("broom", "modelr"))
```

Then run the following R code to load the data set:

```
library(tidyverse)

gorilla_data <- read_csv("http://catalinamedina.github.io/bayesrules-hw-data/gorilla_data.csv")
```

Q1.1

1 Point

Fit a linear model to regress `chest_beat_rate` on `back_breadth`, saving your model as `gorilla_model`. Use `broom::tidy(gorilla_model)` to summarize your model.

Upload a picture of both your code to fit the model and the model results.

 Please select file(s)

Select file(s)

Save Answer

Q1.2

1 Point

What is the estimate of the intercept? Round to two decimal places.

Save Answer

Q1.3

1 Point

What is the estimate of the slope? Round to two decimal places.

Save Answer

Q1.4**1 Point**

We want the 95% confidence interval for the **slope**.

Provide the confidence interval lower bound rounded to two decimal places.

Provide the confidence interval upper bound rounded to two decimal places.

[Save Answer](#)**Q1.5****1 Point**

Provide an interpretation of the 95% confidence interval for the slope in context.

[Save Answer](#)

Q1.6**1 Point**

What is the predicted gorilla chest beat rate for a gorilla with a back breath of 58 centimeters? Round your answer to two decimal places.

[Save Answer](#)**Q1.7****1 Point**

A gorilla in the data set has a back breath of 58 centimeters and a chest beat rate of 1.8. What is the residual for this data point, rounded to two decimal places?

[Save Answer](#)**Q1.8****1 Point**

Use `broom::glance(gorilla_model)` on your model and upload a picture of the output.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q1.9**1 Point**

Identify the R^2 value for your model and provide an interpretation in context.

Save Answer

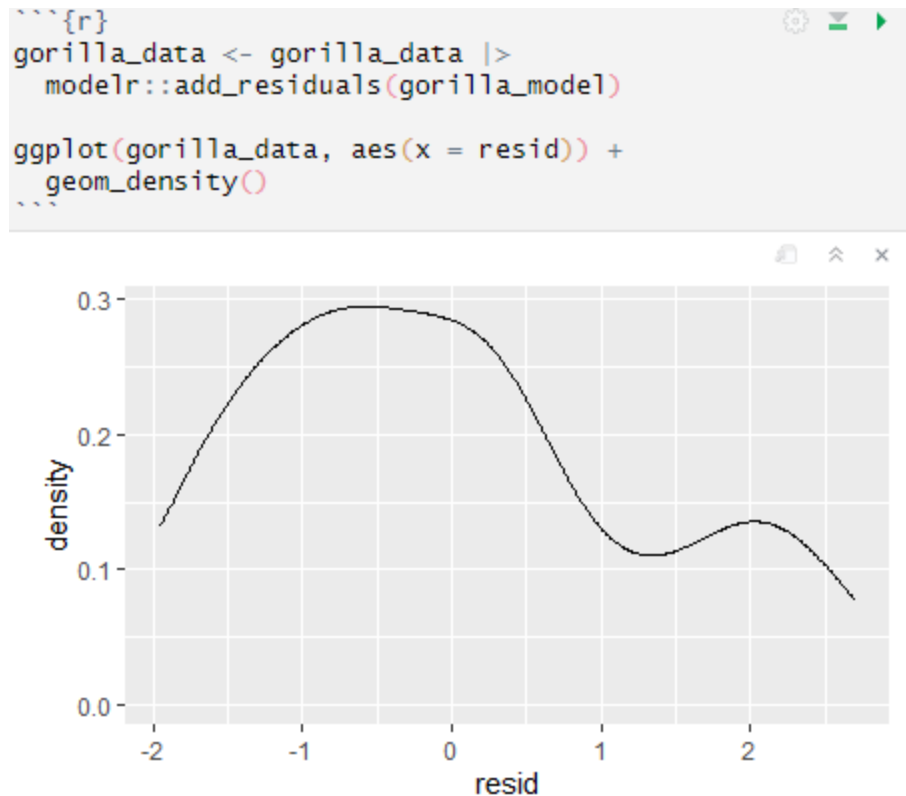
Q1.10**1 Point**

Use the `modelr::rmse(gorilla_model, data = gorilla_data)` to compute the root mean squared error. Round your answer to two decimal places.

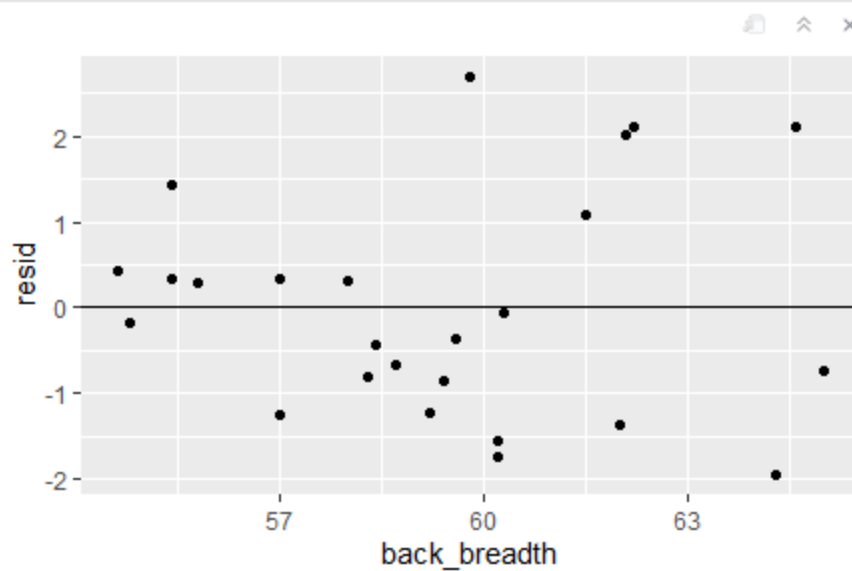
Save Answer

Q2**4 Points**

Now we are going to check the model conditions to make sure that our regression model was appropriate for studying the linear relationship between our response and explanatory variables. Below are a few plots to help with checking these conditions.




```
{r}  
ggplot(gorilla_data, aes(x = back_breadth, y = resid)) +  
  geom_point() +  
  geom_hline(yintercept = 0)
```

**Q2.1**

1 Point

Is the linearity condition met? Explain.

Save Answer**Q2.2**

1 Point

Is the normality condition met? Explain.

Save Answer

Q2.3**1 Point**

Is the constant variance assumption met? Explain.

[Save Answer](#)**Q2.4****1 Point**

Is the independence assumption met? Explain.

[Save Answer](#)

Q3**2 Points**

Silverback gorillas should be full grown by the time they are 20 years old. Maybe the relationship between chest beat rate and back breadth differs for growing versus full grown gorillas. We are going to add another predictor, `age20`, to our model.

Q3.1**1 Point**

Fit a linear model to regress `chest_beat_rate` on `back_breadth` and `age20`, saving your new model as `gorilla_model2`. Use `broom::tidy(gorilla_model2)` to summarize your model. Upload a picture of both your code to fit the model and the model results.

 Please select file(s)**Select file(s)****Save Answer****Q3.2****1 Point**

Which model explains the variation in chest beat rate better? Explain your answer with an appropriate model evaluation metric.

Save Answer

Q4**2 Points**

Let's check our understanding about a few other regression topics discussed in class.

Q4.1**1 Point**

When model conditions are not met, there are some strategies that might help, such as transforming your response variable. Select the scenario where transforming your response variable would definitely **not** be helpful.

When there is a nonlinear relationship between your response and predictor(s)

When your residuals are not normally distributed

When there is a pattern in the residuals, such as non constant variance

When your data are correlated

Save Answer

Q4.2**1 Point**

What strategy can help you avoid overfitting your data?

Compare the R^2 value for each model you are considering

Compare the root mean squared error value for each model you are considering

Split the data into training and testing data

Fit a model with all of the variables and keep the predictors with the lowest p-values

Save Answer

Q5**3 Points**

We are now moving onto questions pertaining to maximum likelihood estimation. For each of the following scenarios, specify if we are investigating the probability or likelihood.

Q5.1**1 Point**

The percentage of vegetarians in a random sample of 100 UCI students was 18%. What is the chance that 25% or more are vegetarians among all anteaters?

probability

likelihood

Save Answer

Q5.2**1 Point**

If the number of lightnings during one hour of storm follows a Poisson Distribution with $\lambda = 5$ what is the chance that 4 lightnings fall in 30 minutes?

probability

likelihood

Save Answer

Q5.3**1 Point**

If the sample average is 50, what is the best estimate that we can make for μ in a Normal Distribution with standard deviation 1?

probability

likelihood

[Save Answer](#)**Q6****5 Points**

Let X be a random variable with the pdf

$$f(x) = \theta x^{\theta-1} \text{ where } 0 < x < 1 \text{ and } \theta > 0$$

Let $\{0.1, 0.2, 0.3\}$ be the three observed values of X . Find the maximum likelihood estimate of θ . Round to two decimal places.

Upload a picture of your work:

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q7**1 Point**

The following two questions are related to the homework choice you made in choosing to do this assignment instead of the other two options. There are no correct answers, this question is only being graded for completeness (note that answering "I felt like it" or "it was interesting" is an insufficient response).

Please explain in detail what led you to choose this topic, gorillas, to investigate.

What was it about the other two options (bikeshare and 2003 movies that led you not to choose them? Please explain in detail.

Save Answer

Q8**1 Point**

- Have you done the homework by yourself, with a peer(s), or used tools like ChatGPT? If you do not acknowledge any help from a person or ChatGPT, we will consider this cheating.

- If you provided help to a peer(s) and/or got help from your peers(s) please explain briefly what this entailed.

- If you used ChatGPT, include the prompts you used and explain how ChatGPT supported your learning.

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